INCH-POUND
MIL-DTL-24231/19E(SH)
26 April 2013
SUPERSEDING
MIL-C-24231/19D(SH)
15 May 1987

DETAIL SPECIFICATION SHEET

CONNECTOR, SINGLE CABLE, TYPE III, 65-CONDUCTOR

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24231.

REQUIREMENTS:

1. Dimensions and configuration: see figures 1 through 7.

2. The gland nut shall be tightened until the packing tightens against the wires. This can be determined by moving the wires gently back and forth until the wire is seized by the packing. The gland nut shall be turned approximately one-fourth of a turn more to hold the wires in place. Do not overtighten the gland nut or attempt to tighten down metal to metal, as this can damage the wires.

3. Refer to MIL-DTL-24231/5 for alternate method of manufacturing connector body and installing guide pin M24231/5-071.

4. Welding of the pin is allowed after the hull fitting body has been age-hardened to achieve its required physical properties.

5. The retainer plates and retainer plate assemblies shall be manufactured from nickel-copper alloy in accordance with Class A, Form 1, hot-finished condition of QQ-N-281 with a minimum yield strength of 40,000 lbf/in^2 at 0.2 percent offset, or nickel-copper alloy in accordance with Class A, Form 1, cold-drawn condition of QQ-N-281 with a minimum yield strength of 50,000 lbf/in^2 at 0.2 percent offset.

6. All dimensions are in inches.

7. Tolerances shall be as follows, or as stated:

a. Decimals – ±0.005 inch.

b. Fractions – ±1/64 inch.

c. Angular dimensions – ±½degree.

8. These connectors are for maintenance purposes only.

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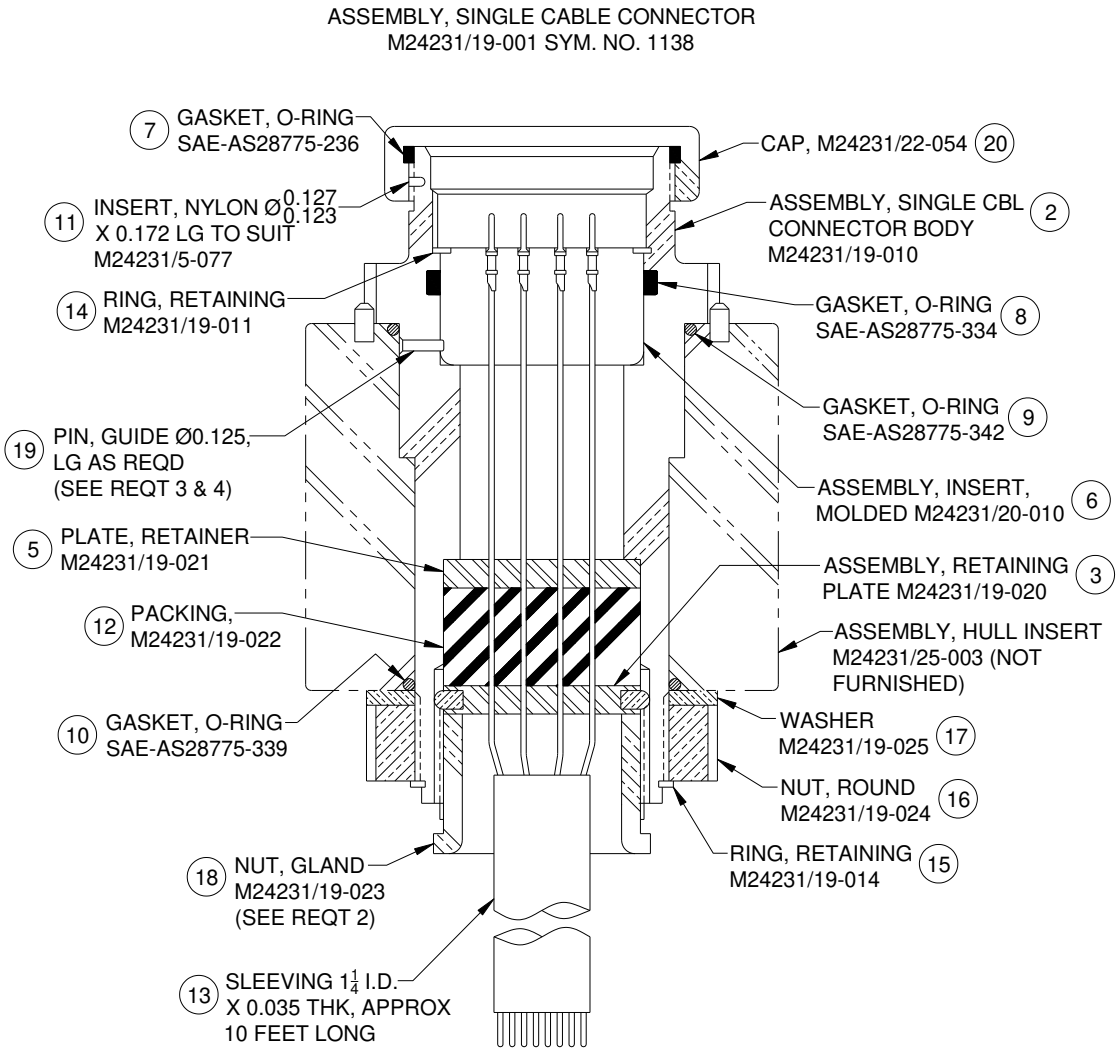


FIGURE 1. Connector, single cable. (1)

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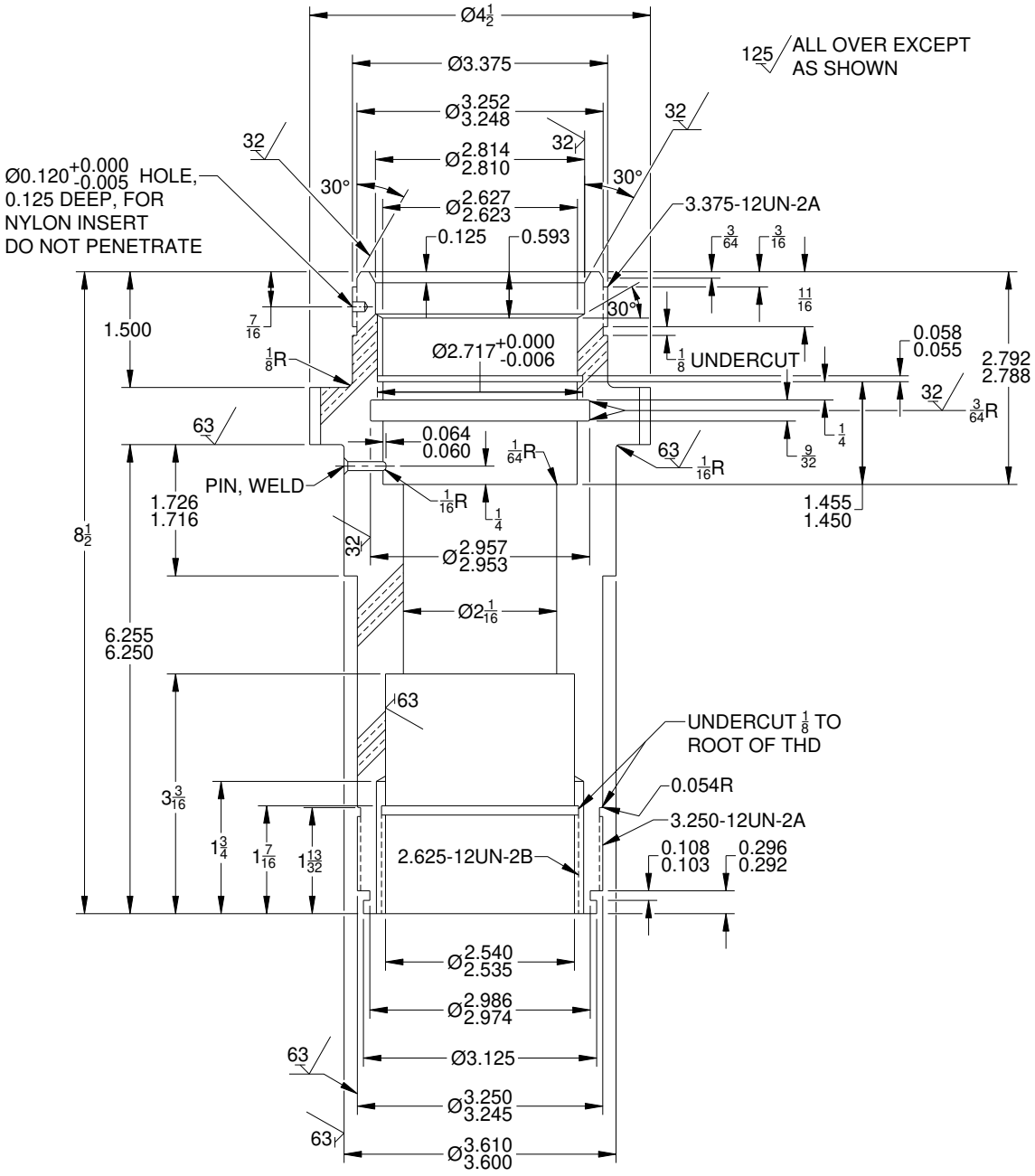


FIGURE 2. Assembly, single cable connector body – M24231/19-010. (2)

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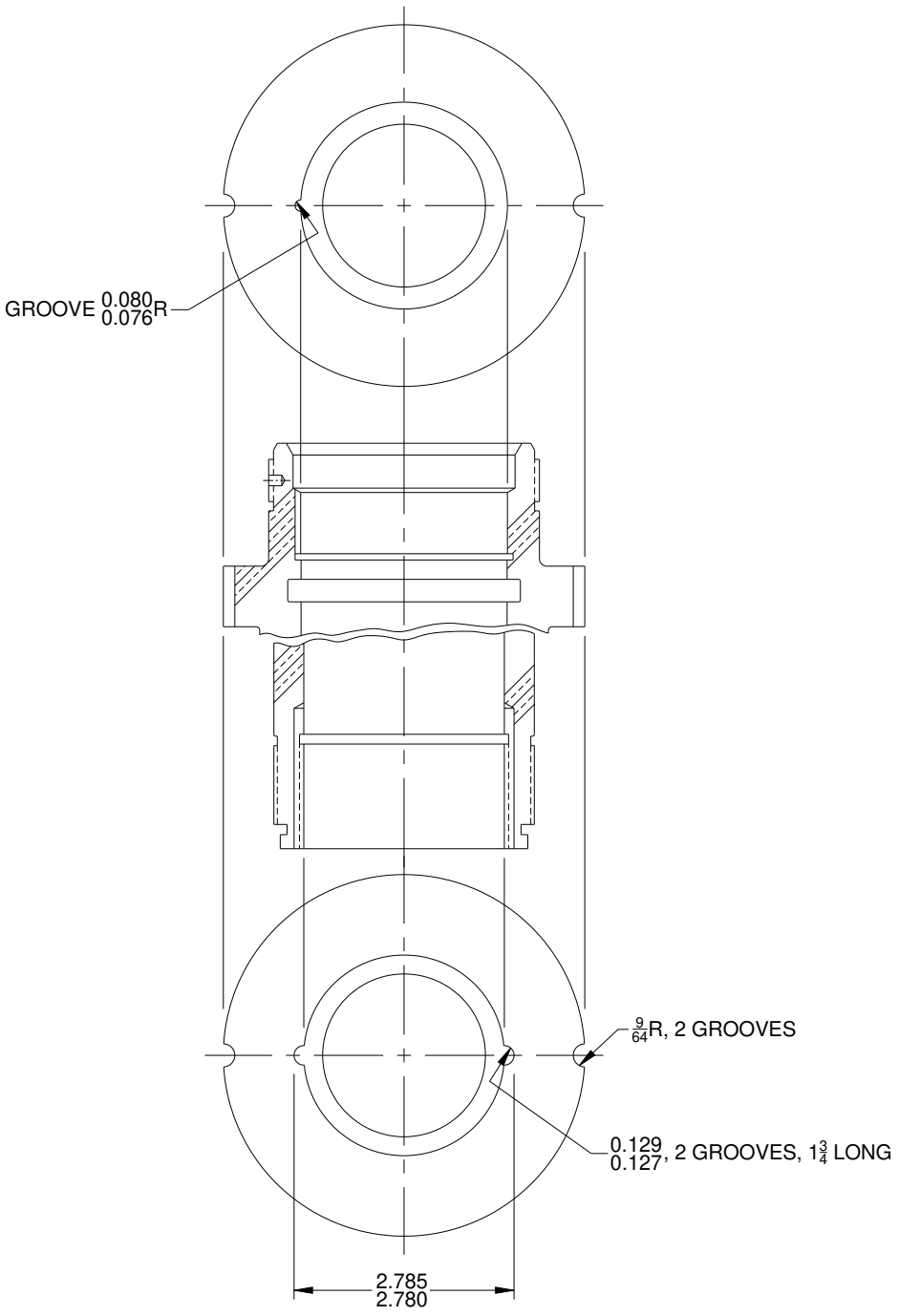


FIGURE 2. Assembly, single cable connector body – M24231/19-010 – Continued.

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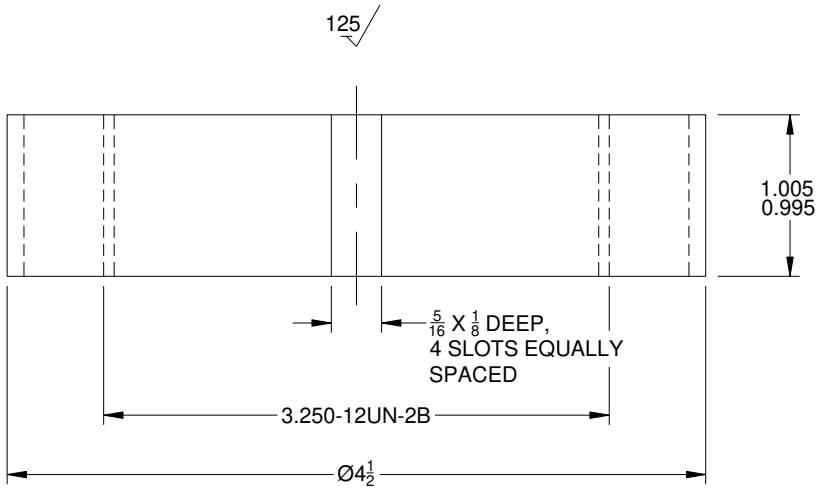


FIGURE 3. Nut, round – M24231/19-024. (16)

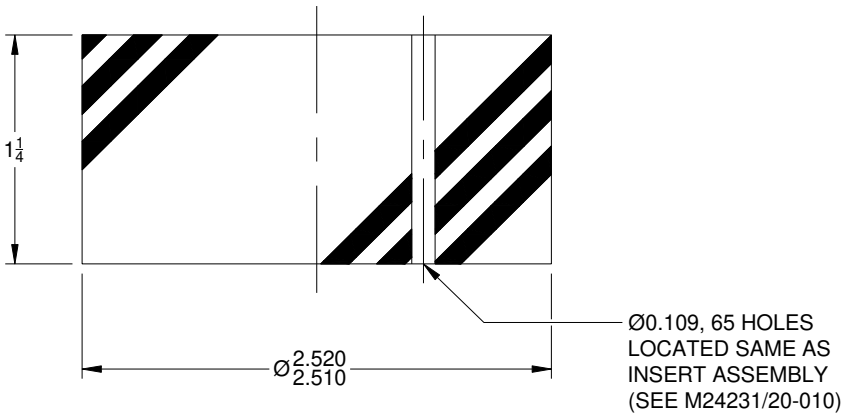


FIGURE 4. Packing – M24231/19-022. (12)

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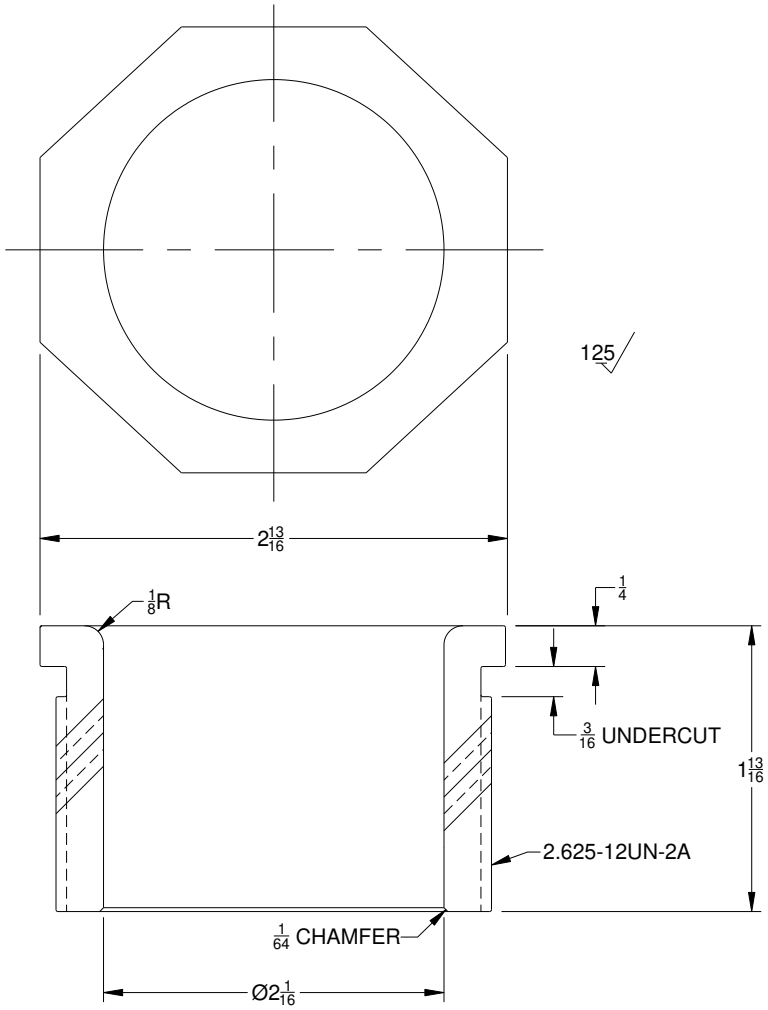


FIGURE 5. Nut, gland – M24231/19-023. (18)

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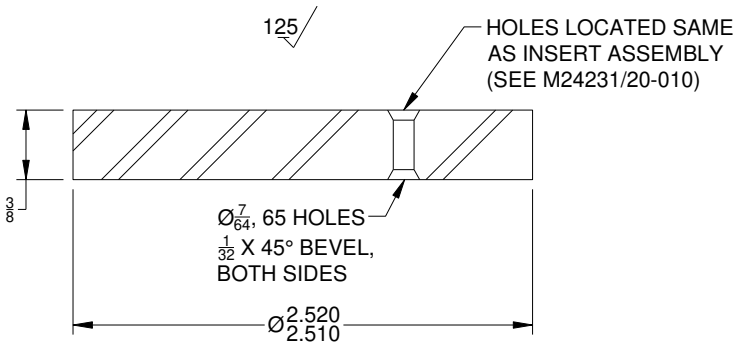


FIGURE 6. Plate, retainer – M24231/19-021. (5)

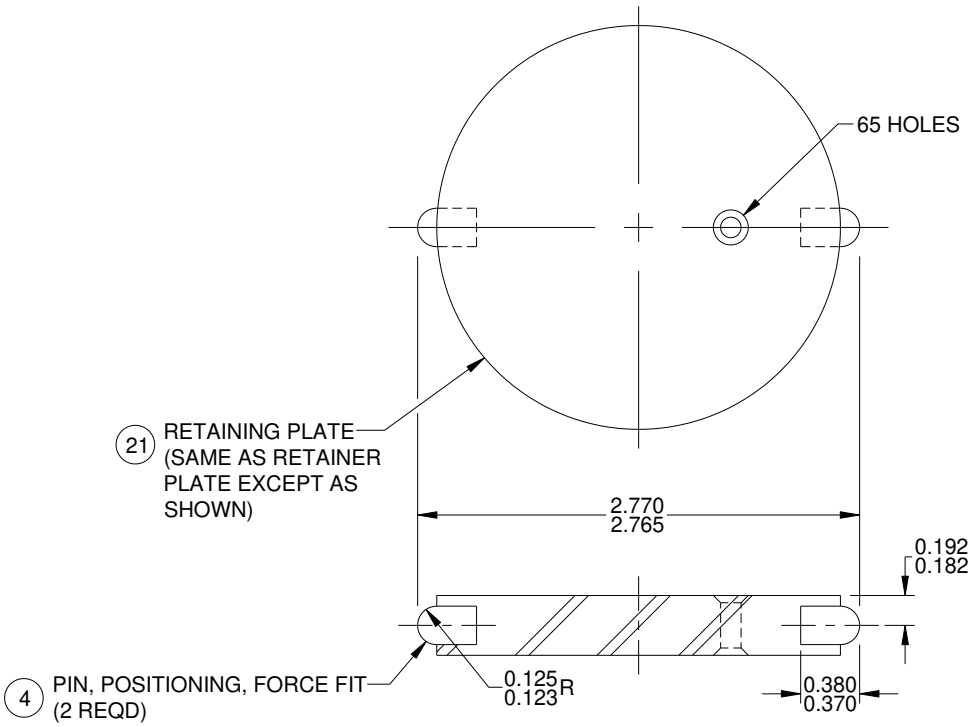


FIGURE 7. Assembly, retaining plate – M24231/19-020. (3)

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TABLE I. Parts list.

Item no. 1/	Description 2/	Quantity for M24231/19- 3/	Material/specification	Remarks
		001		
1	Connector, single cable symbol no. 1138	1	------	See figure 1.
2	Connector body, single cable	1	Ni-Cu-Al, QQ-N-286	See figure 2 and continuation of figure 2.
3	Retaining plate assembly	1	See items 04 and 21.	See figure 7 and requirement 5.
4	Pin, positioning	2	Ni-Cu, Class A, QQ-N-281	See figure 7.
5	Plate, retainer	1	Ni-Cu, Class A, QQ-N-281	See figure 7.
6	Insert assembly, molded	1	M24231/20-010	See figure 1.
7	Gasket, O-ring	1	SAE-AS28775-236 4/	Lubricate with silicone compound, SAE-AS8660; see figure 1.
8	Gasket, O-ring	1	SAE-AS28775-334 4/	See item 7.
9	Gasket, O-ring	1	SAE-AS28775-342 4/	See item 7.
10	Gasket, O-ring	1	SAE-AS28775-339 4/	See item 7.
11	Insert	1	Nylon, Group 1, Class 1, ASTM D4066	See item 7.
12	Packing, Class A, durometer 55±5	1	Rubber, ZZ-R-768	See figure 4.
13	Sleeving, 1¼ id by 0.035 thick	10 ft	Type F, Form U, Category 1, color clear, MIL-I-631	See figure 1.
14	Ring, retaining, M24231/19-011	1	SPIROLOX, RR-262 5/	See figure 1.
15	Ring, retaining, M24231/19-014	1	SPIROLOX, RST-312S 5/	See figure 1.
16	Nut, round	1	Ni-Cu-Al, QQ-N-286	See figure 3.
17	Washer, 4½ od by 3.260 id by 0.187 thick	1	See item 16.	See figure 1.
18	Nut, gland	1	Ni-Al-BRZ, ASTM B150/B150M	See figure 5 and requirement 2.
19	Pin, guide, 0.125 diameter, length to suit	1	Ni-Cu, Class A, QQ-N-281	See requirements 3 and 4.
20	Cap, pressure-proof	1	M24231/22-054	See figure 1.
21	Plate, retaining	1	Ni-Cu, Class A, QQ-N-281	See figure 7 and requirement 5.

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TABLE I. Parts list – Continued.

NOTES:

1/ Item numbers are the same as encircled numbers shown on figures.

2/ See applicable figure or table listed in remarks column for part numbers and dimensions.

3/ Quantity is for one complete assembly of each size.

4/ Nitrile (Buna N); Parker Hannifin Corporation (part number N0756-75), O-ring Division, 2360 Palumbo Drive, Lexington, KY 40509, CAGE 02697, or equal.

5/ Manufactured by Ramsey Corp., St. Louis, MO, or equal.

CHANGES FROM PREVIOUS ISSUE. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Preparing activity:
Navy – SH
(Project 5935-2011-068)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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